87 Prince Street: deck and awning build plans

Project	
Site	87 Prince Street, Mullumbimby NSW, backyard between the studio (trailer) and the shed
Work	4.8 × 3.9 m timber deck with a 5° skillion awning roof (Zincalume over the BBQ end, clear corrugated poly over the rest), and a solid clear-sealed FC wall on the dividing-fence side
Owner	Bryce Noakes
Engineering	Reviewed by **Ollie Harris, structural engineer**: markup of the deck and roof framing plans and a follow-up call, both 2 Nov 2026. Anything he didn't change is approved as drawn. Items from him are tagged engineer.
Issue	P27 · 4 Nov 2026 · supersedes P26: deck stops on the shed-side bearer, cast-in stumps, solid FC wall at the fence end



Live 3D model, drawings and Ollie's marked-up PDF:
tradearts.work/prince-street-deck-awning/p26.html
Spin the model to see any joint.

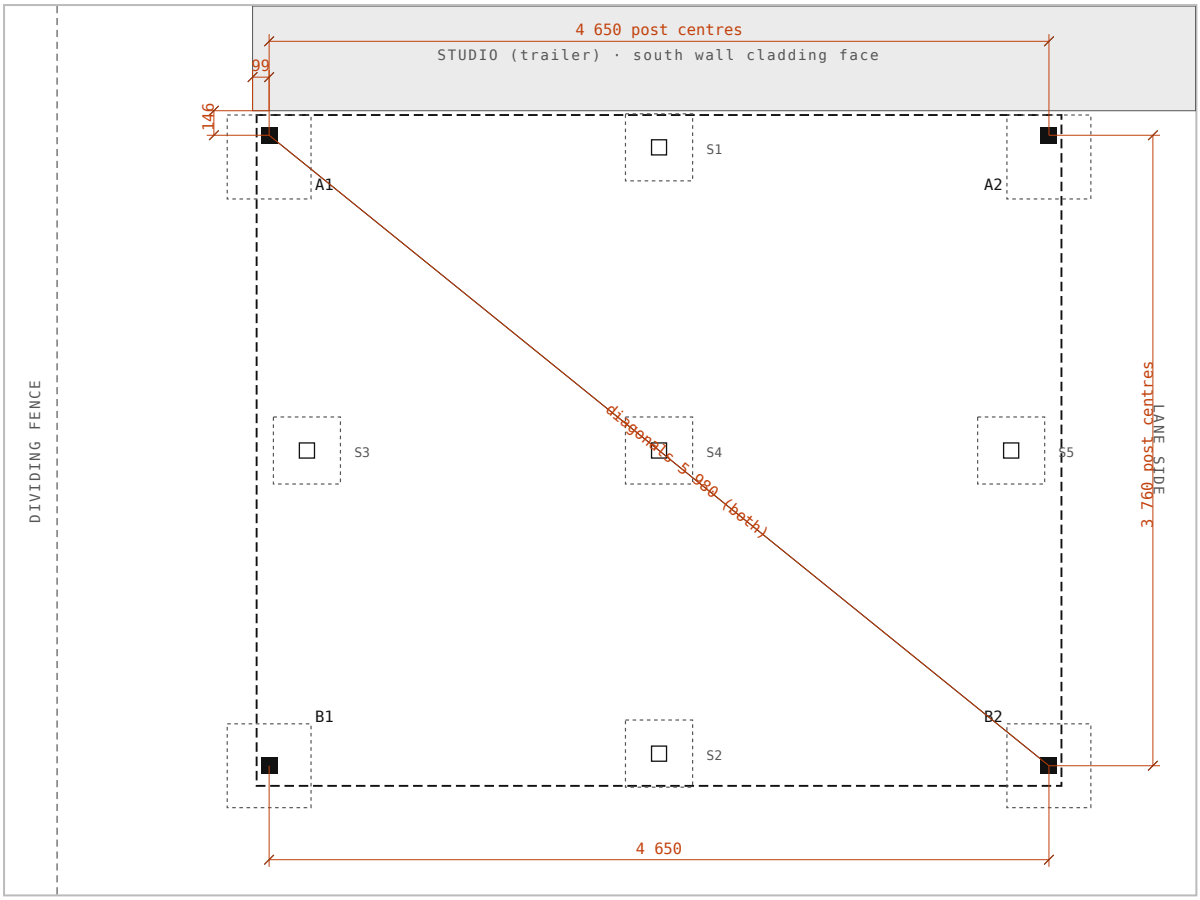
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GENERAL NOTES

- All dimensions in millimetres. Don't scale off the drawings; ask if a dimension is missing.
- Check the setout and both diagonals (5 980) before digging.
- Corner posts: **100×100 hardwood, in-ground durability class**, cast straight into the footings, no brackets engineer. Ollie: a cast-in post must be H4 or an in-ground class hardwood.
- All treated pine H3 above ground; stumps H4. Seal every cut end.
- Bearers and joists run in one continuous length; no joins over supports engineer.
- Bolts M12 hot-dip galvanised with 50×50 square washers under heads and nuts. Bolt counts marked "typical" weren't set by Ollie: two M12 per beam-to-post and bearer-to-post joint, staggered, 2.5 bolt diameters from edges.
- Strap every rafter down to both beams (triple grip or cyclone strap each side).
- Knee braces at all four corner posts engineer. Do a shake test when framed; if a post isn't stiff, add a second brace at the top corners engineer.
- The laundry tower, outdoor kitchen and table in the model are by the owner, not in this scope.

Setout plan



SETOUT PLAN · POST AND STUMP CENTRES · FOOTINGS DASHED

Centres, measured from the studio

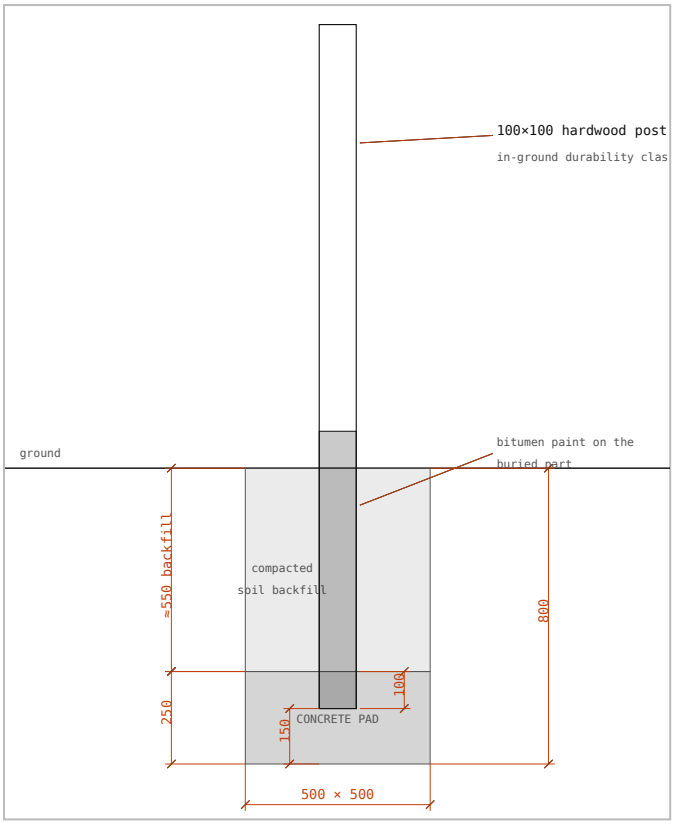
	Along wall from studio west end	Out from cladding	
A1	99	146	corner post
A2	4 749	146	corner post
B1	99	3 906	corner post
B2	4 749	3 906	corner post
S1	2 424	218	stump, line A
S2	2 424	3 834	stump, line B
S3	324	2 026	stump, line C
S4	2 424	2 026	stump, line C
S5	4 524	2 026	stump, line C

Checks

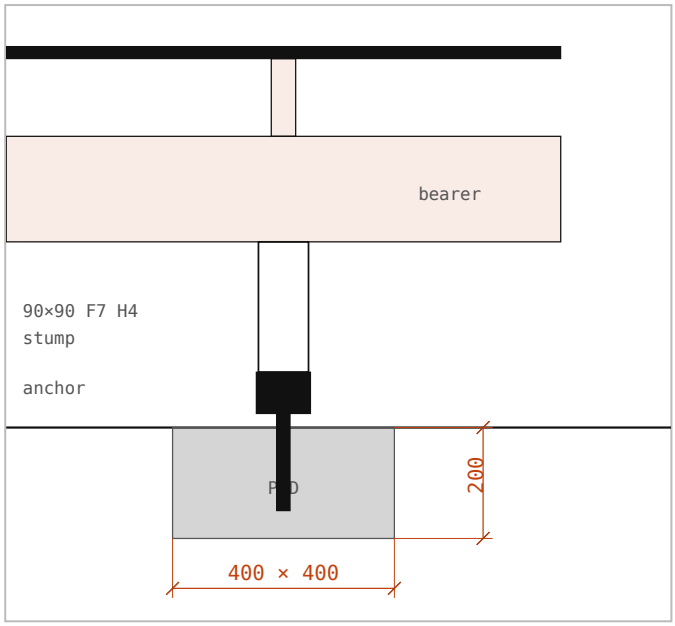
- A1–A2 and B1–B2: 4 650. A1–B1 and A2–B2: 3 760. Both diagonals: 5 980.
- The deck outline is 4 800 × 4 000: 75 past the post centres along the studio, 120 out from the studio-side post centres.
- Line C (S3–S5) is 2 000 out from the deck edge at the studio.

Studio-side holes A1, A2: a 500 hole centred on the post would run about 100 under the studio. Dig them from the studio line outward (hole 0 to 500 out from the deck edge), so the post sits off-centre in the pad, and keep clear of the trailer stands. Check this with Ollie if you're unsure.

Footings



CORNER POST FOOTING · A1, A2, B1, B2



STUMP PAD · S1–S5

Corner posts engineer

1. Dig 500 × 500, **800 deep**, into the clay.
2. Bitumen-paint the buried part of each post.
3. Stand the post in the hole with its foot **150 off the bottom**; plumb it both ways, check centres and diagonals, and prop it so it hangs there.
4. Pour a **250 thick concrete pad**: 150 under the post, post 100 into it. Don't take the post through the concrete.
5. When it has set, backfill the remaining ≈550 with the dug soil, rammed in layers. The soil over the pad is what holds the post down in uplift.
6. Leave the post tops long; cut them flush with the beam tops later.

Stumps engineer

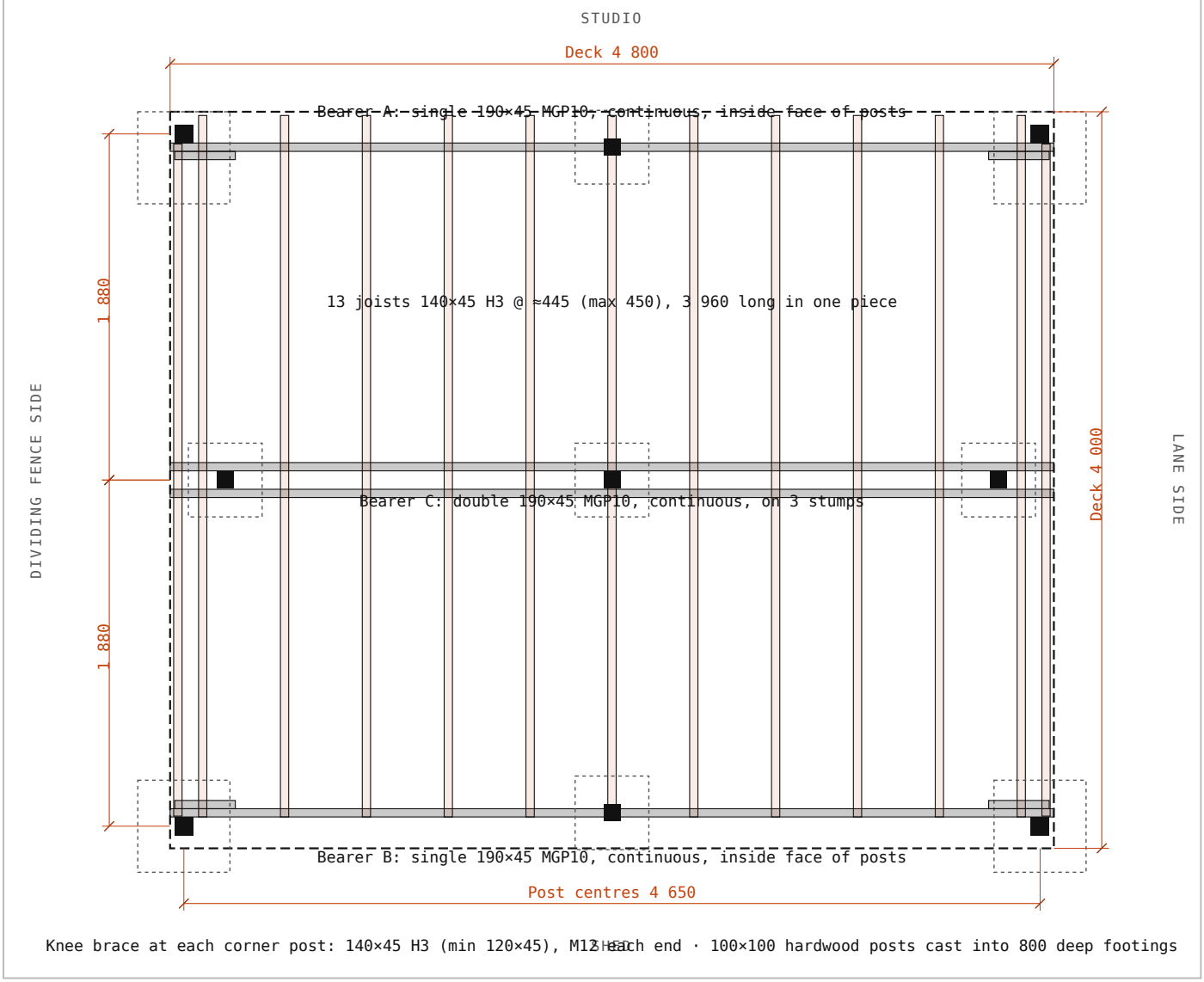
1. Dig 400 × 400, **200 deep**.
2. Hang each 90×90 H4 stump from its bearer, bitumen on the buried end, foot ≈100 off the bottom, and pour so it's ≈100 into the pad. (Replaces the anchors; with concrete under its stem.
3. Line A and B stumps (S1, S2) carry the single bearer on top; line C stumps (S3–S5) sit between the two bearer plies.

Concrete

Corner pads 4 × 0.5 × 0.5 × 0.25	0.25 m³	30 × 20 kg
Stump pads 5 × 0.4 × 0.4 × 0.2	0.16 m³	20 × 20 kg

About 108 bags per m³, plus a few spare.

Deck framing plan



FOOTING AND DECK FRAMING PLAN · LOOKING DOWN · STUDIO AT TOP

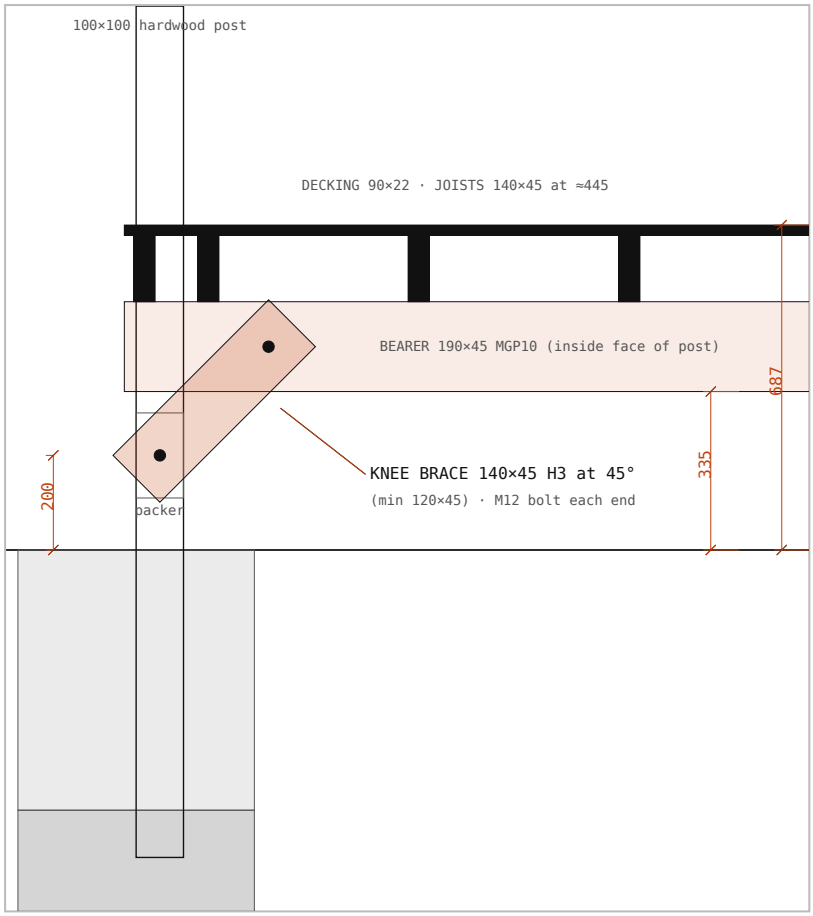
Bearers engineer

- **Lines A and B** (the post lines): a **single 190×45 H3 MGP10**, 4.8 m continuous, bolted to the **inside face** of the posts, sitting on the mid stump.
- **Line C** (middle, 2 000 out): a **double 190×45 H3 MGP10**, 4.8 m continuous, one ply each side of the three stumps, bolted through.
- All bearers continuous and supported by an intermediate post or stump.

Joists and decking

- 13 joists **140×45 H3**, ≈3 810 in one length over all three bearers, ending flush with the outside of bearer B, plus a 140×45 edge board across their ends, at ≈445 centres, max 450 engineer. Ollie wants the bigger joists kept because the laundry and outdoor kitchen sit on this deck.
- Joists set 20 in from each deck edge; edge joists stop at the posts.
- Decking 90×22 H3, 41 rows of full 4.8 m lengths, ≈5 gaps, 20 overhang, notched round the studio-side posts. The deck stops at the shed-side posts, giving the shed steps more room.
- Fix joists to bearers with triple grips (typical).

Corner bracing and bearer details



CORNER POST, BEARER AND KNEE BRACE · ELEVATION ALONG BEARER LINE A (LINE B IS THE MIRROR)

Knee braces engineer

- At **all four corner posts**: a timber brace at 45° from the bottom of the post up to the bearer, in line with the bearer.
- Minimum **120×45**; shown 140×45 H3, ≈465 long.
- **One M12 bolt each end.** Bolt centres 200 above ground on the post and 430 on the bearer (on flat ground).
- The brace runs on the inside face of the bearer, so a 45 packer on the post lines it up (my detail; Ollie only set the brace).
- Shake test each post once framed. If it isn't stiff, add another brace at the top corners engineer.

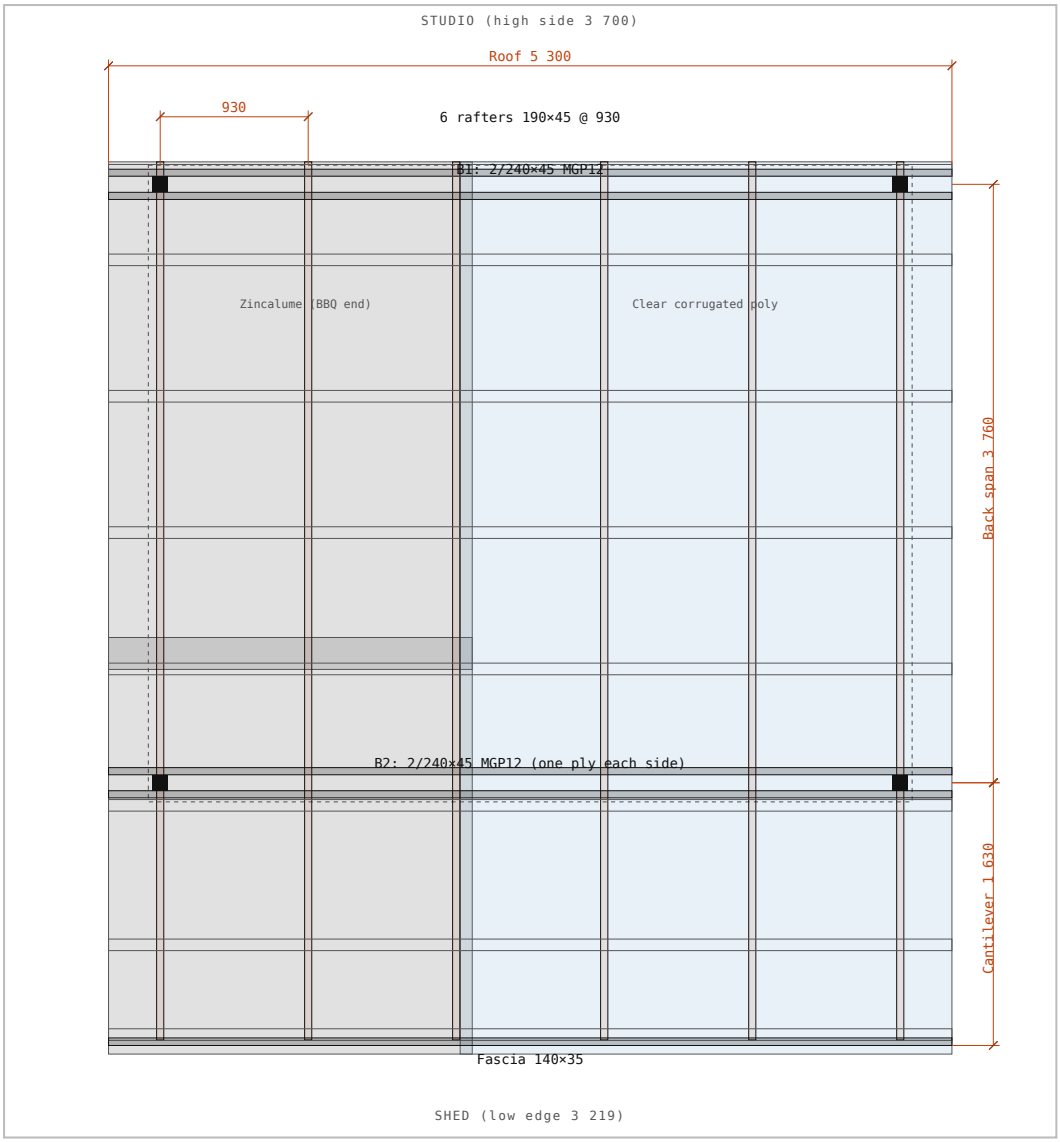
Bearer fixings

- Bearers A and B to each corner post: 2 × M12 through bearer and post (typical).
- S1/S2: stump beside or notched under the single bearer (notch max half the stump), 2 × M12 × 180.
- Bearer C: both plies bolted through each stump, 1–2 × M12 (typical).

Levels

Finished deck top (datum used here)	687 above ground at the posts
Top of joists / bearers	665 / 525
Underside of bearers	335
Knee brace: post end / bearer end (bolt centres)	200 / 430

Roof framing plan



ROOF FRAMING PLAN · LOOKING DOWN · STUDIO AT TOP

Beams engineer

- **B1** (studio side) and **B2** (deck edge, shed side): each **2/240×45 H3 MGP12**, one ply each side of the posts, 5 300 long from 6.0 m.
- Beam tops level along their length; post tops cut flush with the beam tops.
- Bolt each ply pair through the post: 2 × M12 per post (typical).

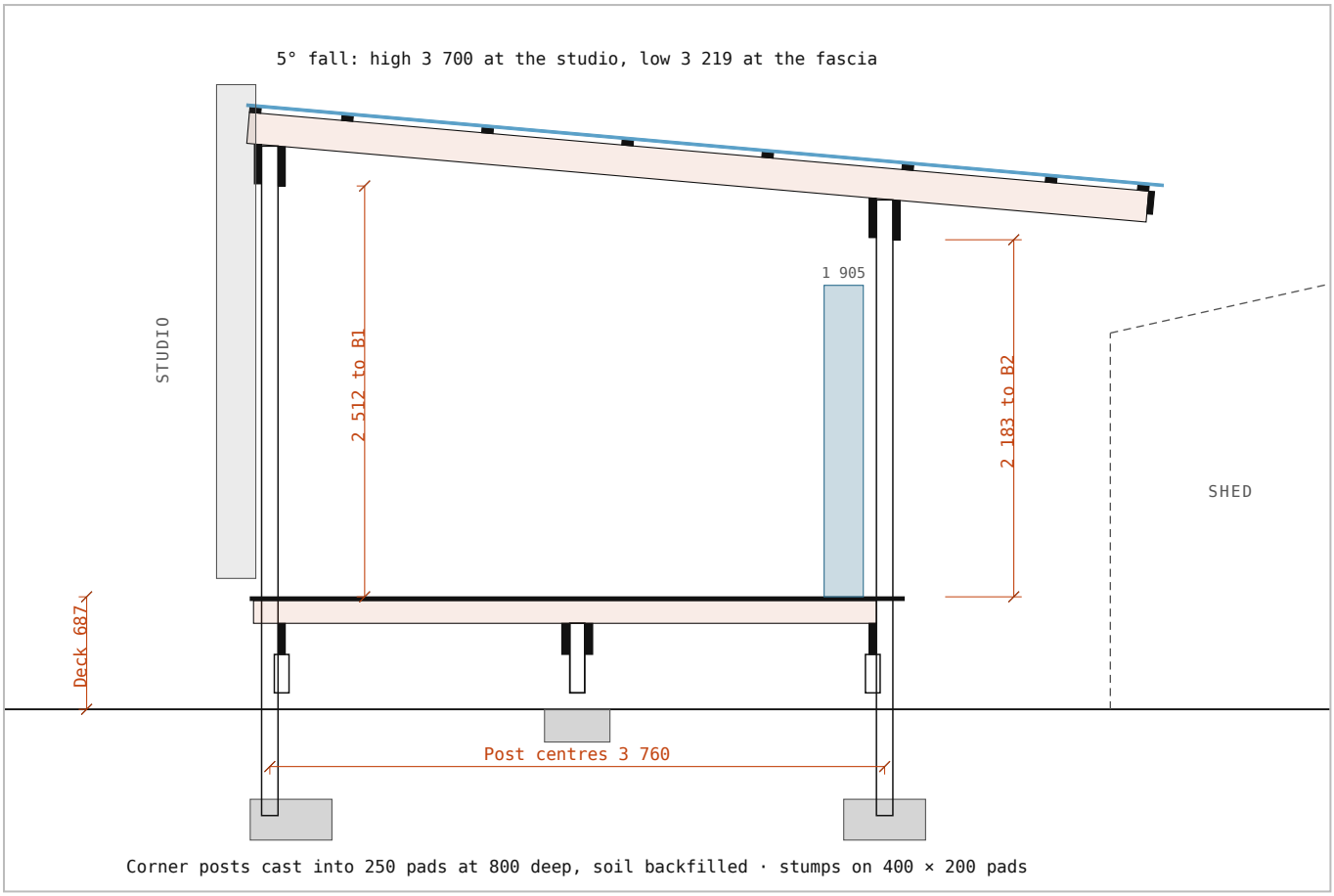
Rafters, approved as drawn

- 6 × **190×45 H3 MGP10** at **930** centres, one over each post line and four between.
- 3 760 back span between the beams, 1 630 cantilever toward the shed; 5° fall away from the studio.
- Seat on both beams (small birdsmouth or packer for the 5°) and strap every rafter to every beam.

Fascia, purlins, sheets

- Fascia **140×35 H3** across the rafter tips (approved).
- 8 purlins 70×35 H3 on flat, full width, 300 past the end rafters.
- Zincalume: 3 runs of 3.2 m sheets over the BBQ (fence) end, lapped 200 over a purlin.
- Clear corrugated poly over the rest, lapped one corrugation over the Zincalume.
- The roof drains onto the shed roof and into the existing shed gutter. No holes in the awning sheets.

Section, levels and beam connection

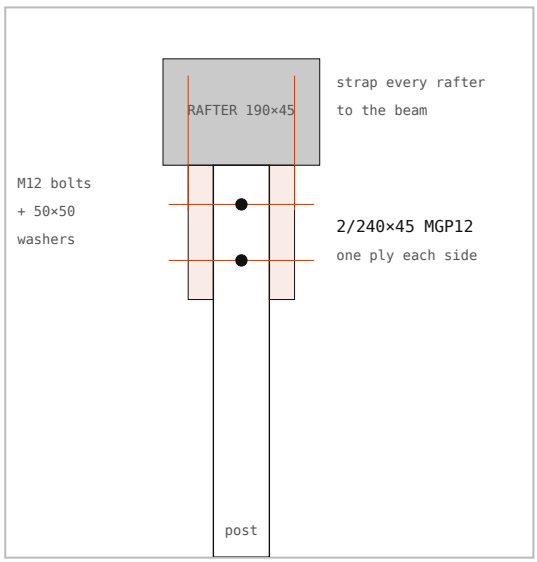


SECTION THROUGH THE DECK AND AWNING, LOOKING TOWARD THE LANE

Levels (above ground at the posts)

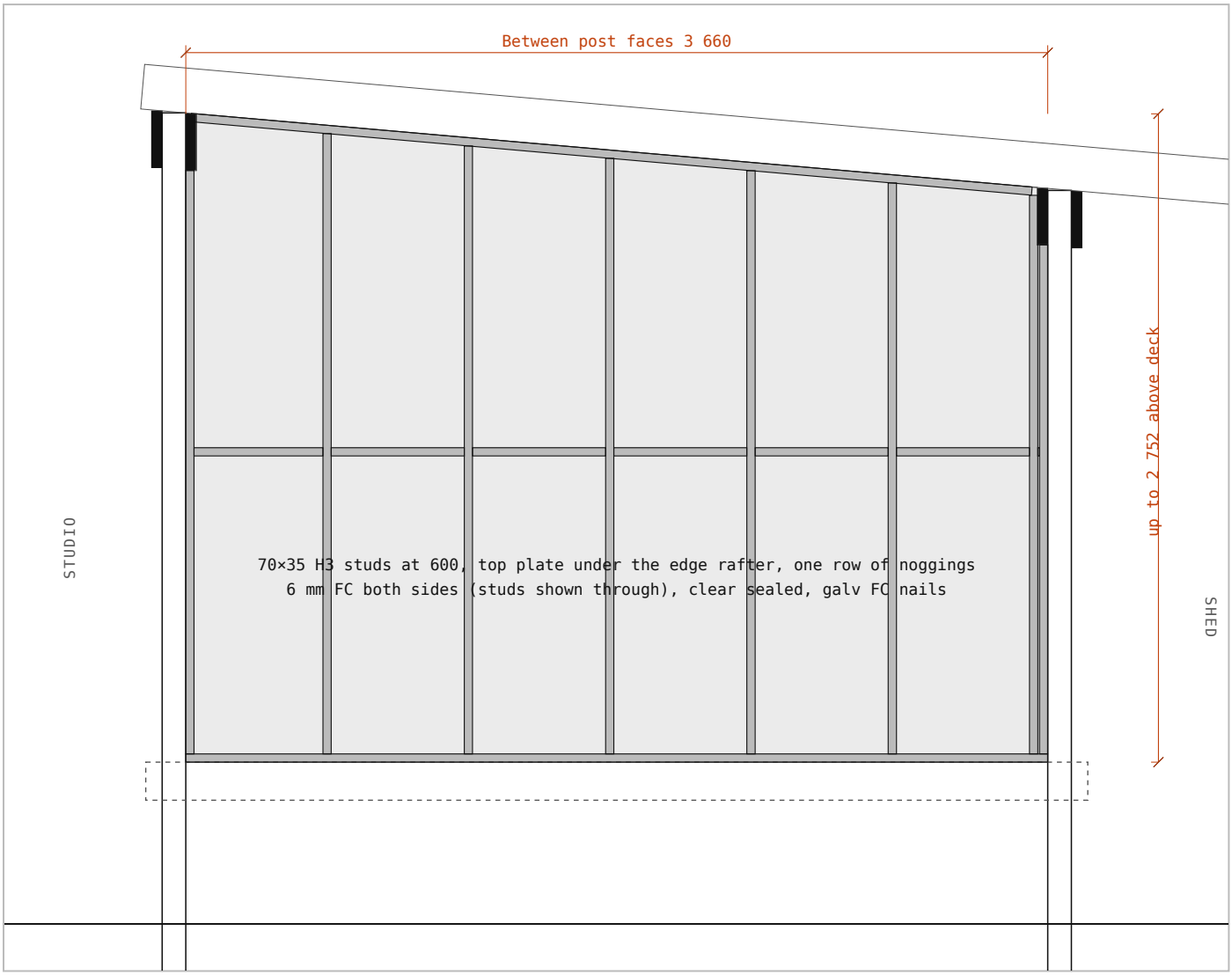
Finished deck top (datum used here)	687 above ground at the posts
Top of joists / bearers	665 / 525
Underside of bearers	335
Knee brace: post end / bearer end (bolt centres)	200 / 430
B2 deck-edge beam: underside / top	2870 / 3114 (2183 clear above the deck)
B1 studio beam: underside / top	3199 / 3443 (2512 above the deck)
Post tops	flush with the top of their beam
Roof	5° fall from the studio to the fascia; fascia underside 2976
Corner post foot	650 below ground (150 above the hole bottom)

Set the deck top on site to suit the studio door threshold; every other level moves with it.



BEAM TO POST · SECTION ACROSS THE BEAM

Fence-end wall



FENCE-END WALL · FROM THE DECK SIDE

- **Frame:** 70×35 H3 studs at 600 between the post faces (≈3 660), bottom plate on the decking, top plate under the edge rafter, one row of noggings at ≈1 300. Every FC edge lands on timber.
- **Height:** 2512 at the studio end (under B1), up to ≈2 750 under the rafter, 2196 at the shed end (under B2).
- The wall is 82 thick (70 + 6 + 6) and sits inside the 100 posts.
- **Sheets:** 6 mm FC both sides, 1 200 × 3 000 stood vertical, 4 per side, vertical joins on studs with a 2–3 gap.
- **Nails:** galvanised FC nails ≈40 × 2.8: 200 at edges, 300 in the field, 12 in from edges, heads flush.
- **Finish:** clear exterior FC sealer on every face and cut edge, ideally before fixing.
- Optional: insulation batts in the cavity for a bit more sound reduction.
- **Text Ollie first:** a solid wall is a bigger wind load on these two posts than the batten screen he checked.
- FC is non-combustible, so the separate BBQ heat panel is no longer needed.

Cutting list and materials

CUTTING LIST

Member	Size	No.	Length
Posts A1, A2 (studio side)	100×100 hardwood, in-ground class	2	4093 (from 4.2 m)
Posts B1, B2 (shed side)	100×100 hardwood, in-ground class	2	3764 (from 3.9 m)
Stumps S1, S2 (lines A and B)	90×90 F7 H4	2	≈600, cast 100 into the pad, bolted to the bearer
Stumps S3–S5 (line C)	90×90 F7 H4	3	≈625, cast 100 into the pad, between the bearer plies
Bearers A and B (single)	190×45 H3 MGP10	2	4 800, one length
Bearer C (double)	190×45 H3 MGP10	2	4 800, one length
Joists	140×45 H3	13	≈3 810, one length (from 4.8 m), to the outside face of bearer B
Deck edge board	140×45 H3	1	4 800, across the joist ends
Knee braces	140×45 H3 (min 120×45)	4	465, 45° ends
Brace packers	140×45 H3 offcut	4	≈180
Decking	90×22 H3	41	4 800 full lengths, ends docked square
B1 and B2 beam plies	240×45 H3 MGP12	4	5 300 (from 6.0 m)
Rafters	190×45 H3 MGP10	6	5521 on the slope (from 6.0 m)
Fascia	140×35 H3	1	5 300
Purlins	70×35 H3, on flat	8	5 300
Wall studs	70×35 H3	8	cut to the rafter underside, ≈2 200–2 750
Wall plates	70×35 H3	2	≈3 660 (top plate follows the rafter)
Wall noggings	70×35 H3	7	≈565
Wall sheets	6 mm FC 1 200 × 3 000	8	4 each side, cut to the rafter line

ORDER LIST

Item	Spec	Qty
Posts	100×100 hardwood, in-ground durability class	2 × 4.2 m, 2 × 3.9 m
Stumps	90×90 F7 H4 (cast in, bitumen on buried ends)	2 × 2.4 m
Bearers	190×45 H3 MGP10 × 4.8 m	4
Joists + edge board	140×45 H3 × 4.8 m	14
Knee braces + packers	140×45 H3 × 4.8 m	1
Decking	90×22 H3 × 4.8 m	44
Roof beams B1, B2	240×45 H3 MGP12 × 6.0 m	4
Rafters	190×45 H3 MGP10 × 6.0 m	6
Fascia	140×35 H3 × 6.0 m	1
Purlins	70×35 H3 × 6.0 m	8
Wall framing	70×35 H3: ≈10 × 3.0 m + 2 × 4.2 m	12
Wall sheets	6 mm FC 1 200 × 3 000	8
Wall nails + sealer	Galv FC nails 40 × 2.8, clear exterior FC sealer	2 kg, ≈8 L
Bolts	M12 HDG bolts/rod, nuts, 50×50 square washers (incl. 8 for the braces)	1 lot
Straps and screws	Triple grips / cyclone straps, batten and decking screws	1 lot
Concrete	20 kg bags: 30 corner pads, 20 stump pads	50
Bitumen paint	For the buried post ends (already bought)	—
End sealer	For cut ends of treated timber	1

Roof sheets ordered separately. Lengths are from the model on level ground; measure post, stump and slat lengths on site before cutting.

Build sequence and checks

1. **Setout** (sheet 2). Peg post and stump centres; check 4 650, 3 760 and both diagonals 5 980.
2. **Dig** the four corner holes 500 × 500 × 800 (A holes from the studio line outward) and five stump holes 400 × 400 × 200.
3. **Posts**: bitumen the buried ends; hang each post plumb, foot 150 off the bottom, propped both ways. Re-check centres and diagonals.
4. **Pour** the 250 corner pads (post 100 in). Stump pads later, with the stumps hung from the bearers.
5. **Hold point:** let the concrete set before loading the posts.
6. **Backfill** ≈550 of soil over each corner pad, rammed in layers.
7. **Stumps**: bolt to the bearers, hang ≈100 off the hole bottom, pour the 400 × 400 × 200 pads.
8. **Bearers**: single on A and B (inside face of posts), double on C; one length each, level, bolted.
9. **Knee braces** at all four corners, M12 each end.
10. **Joists**: 13 at ≈445 (max 450), one length, triple grips.
11. **Roof beams** B1 and B2, 2/240×45 MGP12 each side of the posts; cut post tops flush.
12. **Shake test** each post engineer. If it isn't stiff, add top corner braces before going on.
13. **Rafters** 6 at 930, strapped to both beams; then **fascia** and **purlins**.
14. **Roof sheets**: Zincalume at the BBQ end, clear poly over the rest.
15. **Decking**, notched round the posts.
16. **Fence-end wall**: frame, clad both sides in FC, clear seal.
17. **Seal** all cut ends.

Engineer's changes in this set (Ollie Harris, 2 Nov 2026): single continuous bearers on A and B, double on C; joists continuous, max 450 centres, keep the bigger size; knee braces at all four corners, min 120×45, M12 each end; B1 and B2 2/240×45 MGP12; corner posts 100×100 in-ground hardwood cast straight in: 800 hole, 250 pad with 150 under the post, ≈550 soil backfill; stump pads 400 square × 200. Rafters and fascia approved as drawn.

Questions: Bryce first. Anything structural that isn't on these sheets goes to Ollie Harris, structural engineer.